

ASCII MASTER FLOW – PANEL 1/12: Concept firewall

DESERT -> climatic region

ARIDITY -> long-term moisture deficit

DROUGHT -> temporary anomaly

DESERTIFICATION -> degradation within drylands

MUST REMEMBER: Arid geomorphology links moisture deficit, sparse cover, mechanical weathering, episodic runoff, deflation, abrasion and dune migration; pediment-bajada-playa and Thar land-use pressures frame desertification as land degradation.

ASCII MASTER FLOW – PANEL 2/12: Desert-forming controls

SUBTROPICAL HIGH -> descending dry air

CONTINENTALITY / RAIN SHADOW -> moisture loss

COLD CURRENT -> stable dry coastal air

POLAR COLD -> low absolute moisture

ASCII MASTER FLOW – PANEL 3/12: Aeolian transport

CREEP -> coarse grains roll or slide

SALTATION -> sand grains bounce

SUSPENSION -> fine dust travels far

DEPOSITION -> velocity or carrying power falls

ASCII MASTER FLOW – PANEL 4/12: Dune discriminator

BARCHAN -> limited sand, horns downwind

TRANSVERSE -> abundant sand, one dominant wind

LONGITUDINAL -> aligned with resultant winds

PARABOLIC / STAR -> vegetation anchor / multidirectional winds

ASCII MASTER FLOW – PANEL 5/12: Water in the desert

STORM RUNOFF -> wadi discharge

MOUNTAIN FRONT -> alluvial fan

FANS COALESCE -> bajada

CLOSED BASIN -> playa and salt concentration

ASCII MASTER FLOW – PANEL 6/12: Thar spatial anatomy

WEST -> sandy Marusthali core

EAST -> Bagar and Aravalli transition

SOUTHWEST -> Luni and saline-basin route

NORTH -> canal-alluvial transformation

ASCII MASTER FLOW – PANEL 7/12: Susceptibility-to-degradation

VARIABLE RAIN + WIND -> natural stress

COVER LOSS + SOIL DISTURBANCE -> exposure

WATER MISMANAGEMENT -> salinity or depletion

THRESHOLD CROSSED -> recovery capacity declines

ASCII MASTER FLOW – PANEL 8/12: Aravalli boundary

RELIEF -> modifies local wind and runoff

DRAINAGE -> divides and routes water

HABITAT -> supports connectivity

LIMIT -> not an impermeable desert wall

ASCII MASTER FLOW – PANEL 9/12: Canal balance

BENEFIT -> water supply and production

SEEPAGE -> water-table rise

POOR DRAINAGE -> waterlogging and salinity

MANAGEMENT -> crop-water fit + drainage + monitoring

ASCII MASTER FLOW – PANEL 10/12: LDN response hierarchy

AVOID -> protect functioning land

REDUCE -> improve grazing, tillage and water use

REVERSE -> restore soil, grass and drainage

BALANCE -> monitor gains and losses transparently

CLOSE DISTINCTION: Desert is a climatic moisture condition, desertification is degradation in drylands, and drought is a temporary anomaly; barchan, transverse, longitudinal and parabolic dunes encode different wind/sediment/vegetation controls.

ASCII MASTER FLOW – PANEL 11/12: Restoration matrix

WIND EROSION -> cover, shelter and dune management

RUNOFF EROSION -> contour and watershed treatment

SALINITY -> drainage and water balance

RANGELAND DECLINE -> native grass + institutions

EVIDENCE LIMIT: Wind is not the sole desert agent and every dune is not freely migrating; grazing, irrigation salinity, groundwater, shelterbelts and rainfall variability have locally different effects, so causal conclusions must be spatially qualified.

ASCII MASTER FLOW – PANEL 12/12: Desertification answer spine

DEFINE -> retain UNCCD dryland boundary

MAP -> locate process and pressure

DIAGNOSE -> susceptibility plus accelerator

RESPOND -> process-matched, monitored and qualified